



# Soknarith (Soki) Sem

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## SUMMARY

Materials engineer with hands-on experience in material characterization (SEM, XRD, DSC/TGA), defect analysis, and process optimization. Skilled in statistical methods (SPC, DOE) and root cause investigation. Seeking entry-level Manufacturing Engineer role to apply materials science expertise to semiconductor assembly, chip & wire processes, and reliability improvement.

## WORK EXPERIENCE

### Research Assistant & Graduate Teaching Assistant

May 2024 to Present

Worcester Polytechnic Institute, Worcester, MA

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling – parallels to defect-driven failures (delamination, cracking, corrosion) in semiconductor packages
- Analyzed structure–property–process relationships across temperature regimes – directly applicable to thermal performance and material integrity in chip & wire assemblies
- Computed temperature-dependent free energies and defect formation energies, applying statistical methods (DOS, free energy calculations) to predict material reliability
- Assisted instruction in Applied Analytical Methods (ODEs/PDEs) and Metal Additive Manufacturing

### Chemical Process Engineer

May 2023 to May 2024

Knowles Precision Devices, New Bedford, MA

- Led R&D-to-production transition for high-voltage capacitor dielectric fluids – directly relevant to semiconductor assembly materials (epoxies, solders, encapsulants)
- Optimized manufacturing layout and process flow using lean principles and statistical process controls, reducing production costs by 15%
- Authored SOPs and work instructions for material handling and process scale-up, ensuring cross-functional alignment
- Performed root cause analysis on material-related issues using 5 Whys and data-driven methods

### R & D Technician

Mar 2023 to May 2023

Factorial Energy, Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments – experience with moisture-sensitive materials similar to semiconductor epoxies
- Reduced material waste by 20% through inventory system redesign

### Purification Engineer

Jul 2022 to Mar 2023

Eurofins, Framingham, MA

- Purified therapeutic proteins using AKTA Avant systems (95-99% purity) – attention to contamination control relevant to semiconductor cleanliness
- Extended TFF membrane lifetime by 30% through process optimization

## PROJECTS

### Defect Chemistry of RP Phase (La, Nd, Pr)<sub>2</sub>NiO<sub>4+δ</sub> Study

2024 to 2026

- Performed DFT calculations on oxide RP phase perovskites to quantify defect formation energy, density of states, phonon DOS, elastic/dielectric tensors, and defect free energy
- Constructed chemical potential phase diagram to predict stability regimes – methodology transferable to predicting delamination and corrosion in semiconductor packages
- Applied failure analysis concepts to understand defect-mediated degradation mechanisms

### Bio-based Asphalt Molecular Dynamics Study

2020 to 2022

- Performed classical MD simulations (LAMMPS) to characterize temperature-dependent transport behavior
- Extracted viscoelastic properties across multiple temperature regimes to inform material performance under thermal stress

### Carbon Nanotube Purification Study

2019 to 2020

- Designed and implemented purification protocols achieving 95% nanotube homogeneity – parallels to material purity requirements in semiconductor assembly

## EDUCATION

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### **M.S. Materials Science & Engineering**

Worcester Polytechnic Institute, Worcester, MA

*Relevant coursework:* Defect Chemistry, Thermodynamics of Materials, Transport Phenomena, Statistical Process Control

### **B.S. Chemical Engineering & B.S. Applied Mathematics**

University of Rhode Island, Kingston, RI

Minors: Chemistry, Physics

## TECHNICAL SKILLS

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|---------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Characterization:</b>              | SEM, XRD, DSC/TGA, microscopy, thermal analysis                                                                   |
| <b>Statistical Methods:</b>           | SPC, DOE, Gage R&R, 5 Whys, root cause analysis, Lean/Six Sigma concepts                                          |
| <b>Atomistic Simulation:</b>          | DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)                                                                    |
| <b>Semiconductor Assembly:</b>        | Chip & wire processes, die attach, wire bonding (academic exposure); familiar with solders, epoxies, encapsulants |
| <b>Scientific Computing:</b>          | Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX                                                                 |
| <b>Laboratory:</b>                    | Glovebox operation, HPLC, AKTA FPLC, TFF                                                                          |
| <b>Process Engineering:</b>           | Manufacturing optimization, SOP development, technology transfer                                                  |
| <b>Industry Standards (familiar):</b> | MIL-PRF-19500, MIL-PRF-38534, IPC (self-study)                                                                    |